

Summary of problem:

Code:

Output:

Screenshot of Accepted Bar:

Comments of mine:

In my code, first of all I checked the above conditions and then subtracted from the total size of it by the luck of cases or condition. The time complexity is $O(\text{string size})$ and Space Complexity is $O(\text{string size})$.

Problem name: **Repetitions (18)**

Summary of problem:

There will be given a string of uppercase letters and find out the maximum number of substring of same character.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
int main()
{
    str s;
    cin>>s;
    int a=0,b=0;
    for(int i=0;i<s.size();i++)
    {
        for(int j=i;s[i]==s[j];j++)
        {
            a++;
            i=j;
        }
        if(b<a)
        {
            b=a;
        }
        a=0;
    }
    cout<<b;
}
```

Output:

Input: ATTCGGGA	Output: 3
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Screenshot of Accepted Bar:

Task:	Repetitions
Sender:	Arsenic
Submission time:	2023-09-08 08:02:15 +0300
Language:	C++20
Status:	READY
Result:	ACCEPTED

Comments of mine:

First of all I have run a loop over the characters then iterated a nested loop and counted the same characters until the other character is coming. The time complexity is $O(n^2)$ and space complexity is $O(\text{string size})$.

Problem name: Mike and Palindrome (19)

Summary of problem:

There will be given a string and have to determine if the string could be palindrome by changing one character.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define ev(x) x%2==0
#define od(x) x%2==1
#define mod0(x,y) x%y==0
#define yes_or_no cout<<"YES\n":cout<<"NO\n";
#define newl cout<<endl
int main()
{
    str r;
    cin>>r;
    int n=r.size();
    int a=0;
    for(int i=0;i<n/2;i++)
    {
        if(r[i]!=r[n-i-1])
        {
            a++;
        }
    }
    a==1&&ev(n)||a<=1&&od(n)?yes_or_no
}
```

Output:

Input: abccaa abbcca	Output: YES NO
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Screenshot of Accepted Bar:

219832072	Practice: Arsenic33	798A - 30	GNU C++20 (64)	Accepted	15 ms	8 KB	2023-08-21 21:31:12	2023-08-21 21:31:12
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Comments of mine:

First of all I have counted the total number of miss match and then write a condition for even and odd number of string size . The time complexity is $O(n)$ and space complexity is $O(\text{string size})$.

Problem name: Palindrome Reorder (20)

Summary of problem:

There will be given a string and we have to rearrange it in palindromic order . If not possible print “NO SOLUTION”.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define reverse(x) reverse(x.begin(),x.end())
#define newl cout<<endl
int main()
{
    str s,r="";
```

```

cin>>s;
char d;
int a;
int n=s.size();
unordered_map<char,int>str;
for(int i=0; i<n; i++)
{
    str[s[i]]++;
}
int odd=0;
for(auto p:str)
{
    if(p.second%2==1)
    {
        odd++;
        d=p.first;
        a=p.second;
        if(odd>1)
        {
            cout<<"NO SOLUTION\n";
            return 0;
        }
    }
}
for(auto p:str)
{
    if(p.second%2==0)
    {
        for(int i=1; i<=p.second/2; i++)
        {
            r+=p.first;
        }
    }
}
cout<<r;
if(odd==1)
{
    for(int i=1;i<=a;i++)
    {
        cout<<d;
    }
}
reverse(r);
cout<<r<<endl;
}

```

Output:

Input: AAAACACBA	Output: AACABACAA
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Screenshot of Accepted Bar:

Task:	Palindrome Reorder
Sender:	Arsenic
Submission time:	2023-08-21 21:34:16 +0300
Language:	C++20
Status:	READY
Result:	ACCEPTED

Comments of mine:

First of all I have counted the characters and checked if there is any only one odd number of character count. Then printed the first half and then if there is any odd counted character and then printed second half. If odd is greater than 1 then it is impossible. Time complexity is $O(n*m)$ and space complexity is $O(\text{map size})$.

Problem name: **Lovely Palindrome (21)**

Summary of problem:

There will be given a string of digits of size n . we have to print even length of palindrome string using this character.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define reverse(x) reverse(x.begin(),x.end())
#define newl cout<<endl
int main()
{
    str r;
    cin>>r;
    cout<<r;
    reverse(r);
    cout<<r;
}
```

Output:

Input: 10	Output: 1001
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Screenshot of Accepted Bar:

219832594	Practice: Arsenic33	688B - 13	GNU C++20 (64)	Accepted	15 ms	276 KB	2023-08-21 21:37:02	2023-08-21 21:37:02
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Comments of mine:

In this code I have taken the input first then print it and made it reverse using library function and again printed it. Time complexity is $O(n)$ and space complexity is $O(\text{string size})$.

Problem name: **Palindrome Number (22)**

Summary of problem:

There will be given a string of digits of size n . We have to find out a nearest palindrome number , greater than given and print the subtraction from it and subtracted value must be n digits.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
int main()
{
    int t;
    cin>>t;
    while(t--)
    {
        int n;
```

```

cin>>n;
str r;
cin>>r;
deque<char>dq;
if(r[0]!='9')
{
    for(int i=0;i<n;i++)
    {
        cout<<'9'-r[i];
    }
}
else
{
    int b=0;
    for(int i=n-1;i>=0;i--)
    {
        int a=(r[i]-'0');
        a+=b;
        if(a>1)
        {
            int y=11-a;
            b=1;
            dq.P_F(y+'0');
        }
        else
        {
            int y=1-a;
            b=0;
            dq.P_F(y+'0');
        }
    }
    for(auto it:dq)
    {
        cout<<it;
    }
}
endl;
}
}

```

Output:

Input:	Output:
3	32
2	8646
99	604
4	
1023	
3	
385	

Screenshot of Accepted Bar:

219963266	Practice: Arsenic33	1700B - 39	GNU C++20 (64)	Accepted	31 ms	220 KB	2023-08-22 23:31:17	2023-08-22 23:31:17
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Comments of mine:

First of all I have checked the first character is 9 or not. If it is not 9 then I have made a section of it like all the digits are 9 of size n and if it is 9 then made an another section of it like all the digits of 1 of size (n+1). Time complexity and space complexity is O(n).

Problem name: **Creating String (23)**

Summary of problem:

There will be given a string and print the total possible new string from that string

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define A_sort(x) sort(x.begin(),x.end())
#define vary_perm(x) next_permutation(x.begin(),x.end())
#define newl cout<<endl
int main()
{
    str r;
    cin>>r;
    A_sort(r);
    map<str,int>mp;
    do
    {
        mp[r]++;
    }
    while(vary_perm(r));
    cout<<mp.size();newl;
    for(auto it:mp)
    {
        cout<<it.first;
        newl;
    }
}
```

Output:

Input: abc	Output: 6 abc acb bac bca cab cba
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Screenshot of Accepted Bar:

Task:	Creating Strings
Sender:	Arsenic
Submission time:	2023-08-26 08:14:31 +0300
Language:	C++20
Status:	READY
Result:	ACCEPTED

Comments of mine:

In this code I have used a built in function 'next permutation' to generate all the new string and stored it in a map and printed it. Time complexity is $O(n)$ and space complexity is $O(\text{map size})$.

Problem name: **Digit Summation (24)**

Summary of problem:

There will be given 2 integers and we have to print the sum of last digit of that numbers.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
int main()
{
    str r,s;
    cin>>r>>s;
    int x=r.size()-1,y=s.size()-1;
    cout<<(r[x]-'0')+(s[y]-'0');
}
```

Output:

Input: 13 12	Output: 5
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Screenshot of Accepted Bar:

222356835	Practice: Arsenic33	219158F - 15	GNU C++20 (64)	Accepted	15 ms	0 KB	2023-09-07 21:11:42	2023-09-07 21:11:42
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Comments of mine:

I have taken input as string and determine the last digit , made it integer and sum that integers. Time complexity and space complexity is O(1).

Problem name: **Prime from 1 to n (25)**

Summary of problem:

There will be given a positive integer n and print the prime numbers from 1 to n.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define newl cout<<endl
int is_prime(int a)
{
    for(int i=2;i<a;i++)
    {
        if(a%i==0)
        {
            return 0;
        }
    }
    return 1;
}
int main()
{
    int a;cin>>a;
    for(int i=1;i<=a;i++)
    {
        if(i==1||i%2==0&& i!=2||is_prime(i)==0)
        {
            continue;
        }
        else
        {
            cout<<i<<" ";
        }
    }
}
```

```
}
}
```

Output:

Input: 10	Output: 2 3 5 7
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Screenshot of Accepted Bar:

208434218	Practice: Arsenic33	2194321 - 26	GNU C++20 (64)	Accepted	15 ms	0 KB	2023-06-04 17:29:42	2023-06-04 17:29:42
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Comments of mine:

I have made a function which can determine if the number is prime or not . If prime then printed it. Otherwise skipped. Time complexity is (n^2) and space complexity is $O(1)$.

Problem name: **Range Sum (26)**

Summary of problem:

There will given 2 integers and we have to calculate the summation from minimum to maximum number(inclusive).

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
int main()
{
    int t;
    cin>>t;
    while(t--)
    {
        ll a,b;
        cin>>a>>b;
        a<b?a--:b--;
        cout<<abs(b*(b+1)/2-a*(a+1)/2)<<endl;
    }
}
```

Output:

Input: 4 3 6 2 11 4 16 7 17	Output: 18 65 130 132
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Screenshot of Accepted Bar:

208678298	Practice: Arsenic33	326907D - 14	GNU C++20 (64)	Accepted	624 ms	0 KB	2023-06-06 12:39:31	2023-06-06 12:39:32
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Comments of mine:

First of all, I have determined the minimum and maximum number through ternary operator then used the formula of the summation of n natural numbers. The time complexity and Space Complexity is $O(1)$.

Problem name: **Two Intervals (27)**

Summary of problem:

There will be given 4 integers and we have to print the boundaries of their intersection. If it is possible then print the lower to upper boundary otherwise print -1.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
int main()
{
    vector<ll>A(5);
    cin>>A[1]>>A[2]>>A[3]>>A[4];
    if(A[3]<A[1]&&A[4]<A[1]||A[3]>A[2]&&A[4]>A[1])
    {
        cout<<-1;
    }
    else
    {
        A[3]>A[1]?cout<<A[3]:cout<<A[1];cout<<" ";
        A[2]>A[4]?cout<<A[4]:cout<<A[2];
    }
}
```

Output:

Input: 1 15 5 27 2 5 6 12	Output: 5 15 -1
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Screenshot of Accepted Bar:

222108025	Practice: Arsenic33	219158X - 17	GNU C++20 (64)	Accepted	15 ms	0 KB	2023-09-06 19:02:17	2023-09-06 19:02:17
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Comments of mine:

First of all I have determined the cases due to I got -1 and used a vector to store the data. Then I have compared the number 1st to 3rd and 2nd to 4th and print the answer. The time complexity and space complexity is O(1).

Problem name: **Convert to Decimal 2 (28)**

Summary of problem:

There will be given an integer and we have to count the 1 after transforming the integer in binary representation and print the equivalent decimal number of using total 1.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define ev(x) x%2==0
#define od(x) x%2==1
#define newl cout<<endl
int main()
{
    int t;
    cin>>t;
```

```

while(t--)
{
    int a;
    cin>>a;
    int x=0;
    while(a!=0)
    {
        if(od(a))
        {
            x++;
        }
        a=a>>1;
    }
    cout<<(1<<x)-1;
    newl;
}
}

```

Output:

Input:	Output:
3	3
10	7
7	1
8	

Screenshot of Accepted Bar:

220467141	Practice: Arsenic33	219432X - 18	GNU C++20 (64)	Accepted	15 ms	0 KB	2023-08-26 09:11:04	2023-08-26 09:11:04
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Comments of mine:

In this case, I have checked the least significant bit by using right shift method and if it is odd then count it. After that, printed the left shift number of 1 by subtracting 1. The time complexity is $O(\log n)$ and space complexity is $O(1)$.

Problem name: **Hard Compare (29)**

Summary of problem:

There will be given 4 integers a,b,c,d and we have to determine if $a^b > C^d$ then print 'YES' otherwise print 'NO'.

Code:

```

#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define yes_or_no cout<<"YES\n":cout<<"NO\n"
#define newl cout<<endl
int main()
{
    double a,b,c,d;
    cin>>a>>b>>c>>d;
    double x=b*log10(a),y=d*log10(c);
    x>y?yes_or_no;
}

```

Output:

Input:	Output:
3 2 5 4	NO
5 2 4 2	YES

Screenshot of Accepted Bar:

220468195	Practice: Arsenic33	219158Z - 22	GNU C++20 (64)	Accepted	15 ms	12 KB	2023-08-26 09:21:17	2023-08-26 09:21:17
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Comments of mine:

First of all, I have taken the number in double for clear calculation. I have used logarithm method to avoid integer overflow and then calculated the logarithm value of them and compared it as the statement and printed the answer. The time complexity is $O(\log_{10} n)$ and space complexity is $O(1)$.

Problem name: **B-1% (30)**

Summary of problem:

There will be given an integer and we have to determine the years need to equal the money to given integer if the deposit money 100 is increasing interest rate of 1% compounded annually.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define yes_or_no cout<<"YES\n":cout<<"NO\n"
#define newl cout<<endl
int main()
{
    ll a;
    cin>>a;
    ll x=100;
    int ans=0;
    while(x<a)
    {
        x+=(x/100);
        ans++;
    }
    cout<<ans;
}
```

Output:

Input: 1000000000000000000 1333333333	Output: 3760 1706
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Screenshot of Accepted Bar:

2023-09-03 21:07:42	B - 1%	Arsenic33	C++ (GCC 9.2.1)	200	1611 Byte		7 ms	3632 KB	Detail
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Comments of mine:

As the statement , I have run a loop until the fixed value is not equal to given value by adding with the fixed value of interest and counted to calculate the years and print it. The space complexity is $O(1)$.

Problem name: **Theatre Square (31)**

Summary of problem:

There will be given 3 integers. First 2 integers are the size of theatre square and last one is the size of square flagstone. We have to determine the total number of flagstone is required to pave the square.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
```

```

#define str string
#define mod0(x,y) x%y==0
#define newl cout<<endl
int main()
{
    ll n,m,a;
    cin>>n>>m>>a;
    n=mod0(n,a)?n/a:n/a+1;
    m=mod0(m,a)?m/a:m/a+1;
    cout<<n*m;
}

```

Output:

Input: 6 6 4	Output: 4
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Screenshot of Accepted Bar:

222108666	Practice: Arsenic33	1A - 10	GNU C++20 (64)	Accepted	0 ms	8 KB	2023-09-06 19:07:50	2023-09-06 19:07:50
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Comments of mine:

First of all I have checked if the flagstone size is divisible or not. If it is not divisible then add 1 to divide value and do the same thing with second integer of theatre square dimension. After that I have printed the answer. The time complexity and space complexity is $O(1)$.

Problem name: **Even Odds (32)**

Summary of problem:

There will be given 2 integers first is total numbers and second is position of a number. We have to determine the value if we write 1 to n odd integers and then 1 to n even integers.

Code:

```

#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define ev(x) x%2==0
#define od(x) x%2==1
#define mod0(x,y) x%y==0
#define newl cout<<endl
int main()
{
    ll x,y,z;
    cin>>x>>y;
    z=mod0(x,2)?x/2:(x/2)+1;
    z>=y?cout<<(2*y)-1:cout<<2*(y-z);
}

```

Output:

Input: 10 3 7 7	Output: 5 6
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Screenshot of Accepted Bar:

222109638	Practice: Arsenic33	318A - 17	GNU C++20 (64)	Accepted	30 ms	8 KB	2023-09-06 19:16:10	2023-09-06 19:16:10
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Comments of mine:

First of all determine the middle number of that sequence if second one is least equal than mid value then print the odd number. Otherwise, subtracting from mid value by second number and print the even number. The time complexity and space complexity is $O(1)$.